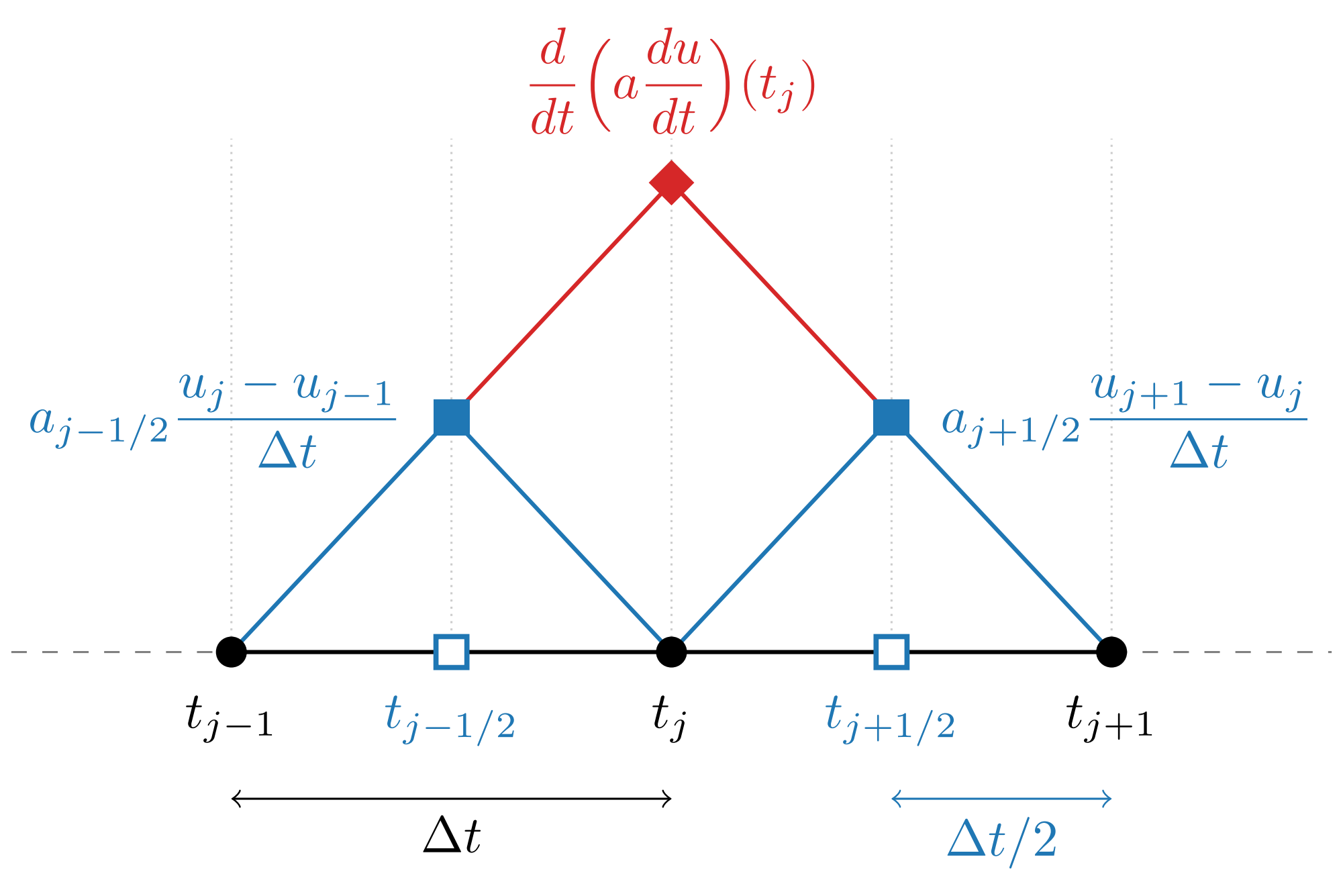

$\frac{d}{dt}\left(a\frac{du}{dt}\right)(t_j)$
$a_{j-1/2}\frac{u_j - u_{j-1}}{\Delta t}$
$a_{j+1/2}\frac{u_{j+1} - u_j}{\Delta t}$
t_{j-1}
$t_{j-1/2}$
t_j
$t_{j+1/2}$
t_{j+1}
Δt
$\Delta t/2$